

Linear Algebra Done Right

1001

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Chapter 6

Inner Product Spaces

6A Inner Products and Norms

1. Inner Products

Def. Dot Product

$$x \cdot y := \sum x_i y_i \quad x, y \in \mathbb{R}^n$$

希望把 $\mathbb{R}^n$ 的点积推广到 $\mathbb{C}^n$ 上。对于 $z = (z_1, \dots, z_n)$, 定义 $\|z\| = \sqrt{\sum |z_i|^2}$ 是自然的。但在 $\mathbb{C}$ 上 $|z_i| = z_i \cdot \bar{z}_i$, 于是 $\|z\|^2 = \sum z_i \cdot \bar{z}_i \Rightarrow \|z\|^2 = z \cdot \bar{z}$ 。这进一步诱导 $\langle u, v \rangle := u \cdot \bar{v}$ 的定义。于是 $\mathbb{R}^n$ 上的对称性变为了 $\mathbb{C}^n$ 上的共轭对称性。

Def. Inner Product

An inner product on V is a function that takes each ordered pair (u, v) of elements of V to a number $\langle u, v \rangle \in \mathbb{F}$ and has the following properties:

- **positivity**

$$\langle v, v \rangle \geq 0 \quad \forall v \in V$$

- **definiteness**

$$\langle v, v \rangle = 0 \Leftrightarrow v = 0$$

- **additivity in first slot**

$$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle \quad \forall u, v, w \in V$$

- **homogeneity in first slot**

$$\langle \lambda u, v \rangle = \lambda \langle u, v \rangle \quad \forall \lambda \in \mathbb{F} \quad \forall u, v \in V$$

- **conjugate symmetry**

$$\langle u, v \rangle = \overline{\langle v, u \rangle} \quad \forall u, v \in V$$

E.g.

- Euclidean inner product $\langle w, z \rangle := \sum w_i \bar{z}_i$
- For $c_1, \dots, c_n > 0$, define $\langle w, z \rangle := \sum c_i w_i \bar{z}_i$
- An inner product can be defined on the vector space of continuous real-valued functions on the interval $[-1, 1]$ by $\langle f, g \rangle := \int_{-1}^1 f g$
- On $\mathcal{P}(\mathbb{R})$, $\langle p, q \rangle := p(0)q(0) + \int_{-1}^1 p' q'$ or $\langle p, q \rangle := \int_0^\infty p(x)q(x) \exp(-x) dx$

Def. Inner Product Space

A vector space V with an inner product.

Prop.

- For each fixed $v \in V$, the function $\langle \cdot, v \rangle : V \rightarrow \mathbb{F}$, $u \mapsto \langle u, v \rangle$ is a linear map.

- $\langle 0, v \rangle = \langle v, 0 \rangle = 0 \quad \forall v \in V$

线性映射映 $0 \mapsto 0$, 故 $\langle 0, v \rangle = 0$ 。共轭对称性继而推出后者。

- $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle \quad \forall u, v, w \in V$

$$\langle u, v + w \rangle = \overline{\langle v + w, u \rangle} = \overline{\langle v, u \rangle} + \overline{\langle w, u \rangle} = \langle u, v \rangle + \langle u, w \rangle$$

d. $\langle u, \lambda v \rangle = \overline{\lambda} \langle u, v \rangle \quad \forall \lambda \in \mathbb{F} \quad \forall u, v \in V$

$$\langle u, \lambda v \rangle = \overline{\langle \lambda v, u \rangle} = \overline{\lambda \langle v, u \rangle} = \overline{\lambda} \overline{\langle v, u \rangle} = \overline{\lambda} \langle u, v \rangle$$

2. Norms

Def. Norm, $\|v\|$

$$\|v\| := \sqrt{\langle v, v \rangle}$$

Prop.

a. $\|v\| = 0 \Leftrightarrow v = 0$

b. $\|\lambda v\| = |\lambda| \|v\| \quad \forall \lambda \in \mathbb{F}$

$$\|\lambda v\|^2 = \langle \lambda v, \lambda v \rangle = \lambda \overline{\lambda} \langle v, v \rangle = |\lambda|^2 \|v\|^2$$

Def. Orthogonal

u, v are called orthogonal iff $\langle u, v \rangle = 0$.

Thm. Pythagorean Theorem

If u and v are orthogonal, then

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2$$

Thm. Cauchy-Schwarz Inequality

$$|\langle u, v \rangle| \leq \|u\| \|v\|$$

This inequality is an equality iff one of them is a scalar multiple of the other.

Thm. Triangle Inequality

$$\|u + v\| \leq \|u\| + \|v\|$$

This inequality is an equality iff one of them is a nonnegative real multiple of the other.

Thm. Parallelogram Equality 平行四边形等式

$$\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2)$$

6B Orthonormal Bases

6C Orthogonal Complements

1. Orthogonal Complement

Def. Orthogonal Complement

$U \subset V$, **not necessary to be a subspace**, then the orthogonal complement of U is the set of all vectors that are orthogonal to every vector in U

$$U^\perp := \{v \in V : \forall u \in U, \langle u, v \rangle = 0\}$$

e.g. 取 V 为所有二维向量的集合, $U = \{(1, 1)\}$, 则 $U^\perp$ 为 $y = -x$ 上的所有向量。

Prop.

a. $U^\perp$ is a subspace of V .

验证定义。

b. $\{0\}^\perp = V$, $V^\perp = \{0\}$

前者显然, 后者 $u \neq 0 \Rightarrow \langle u, u \rangle \neq 0 \Rightarrow u \notin V^\perp$

c. $U \cap U^\perp \subset \{0\}$

显然。不取等因为可以 $0 \notin U$

d. If subset $G \subset H \subset V$, then $H^\perp \subset G^\perp$

显然。

Prop. A natural decomposition

If $U \subset V$ is **finite**-dimensional subspace, then

$$V = U \oplus U^\perp$$

proof. **rewrite using the orthogonal basis of U**

Let $\{e_m\}$ be a orthogonal basis of U , consider

$$v = \underbrace{\sum_{u \in U} \langle v, e_i \rangle e_i}_{u \in U} + \underbrace{v - \sum_{u \in U} \langle v, e_i \rangle e_i}_w$$

Only need to prove $w \in U^\perp$, and thus only need to prove $\langle w, e_i \rangle = 0$ for all e_i .
Obviously.

Cov. If $U \subset V$ is finite-dimensional subspace, then $\dim U^\perp = \dim V - \dim U$

Prop. Orthogonal complement of orthogonal complement

If $U \subset V$ is a finite-dimensional subspace, then

$$U = (U^\perp)^\perp$$

proof.

• $U \subset (U^\perp)^\perp$: by definition, $u \in U \Rightarrow \langle u, v \rangle = 0 \forall v \in U^\perp \Rightarrow u \in (U^\perp)^\perp$

- $(U^\perp)^\perp \subset U$: suppose $v \in (U^\perp)^\perp$. Since $V = U \oplus U^\perp$, then $v = u + w$ for some $u \in U, w \in U^\perp$. From above $U \subset U^{\perp\perp}$, so $u \in (U^\perp)^\perp$. Therefore

$$\left. \begin{array}{l} w = v - u \in (U^\perp)^\perp \\ w \in U^\perp \end{array} \right\} \Rightarrow w \in U^\perp \cap U^{\perp\perp} = \{0\} \Rightarrow w = 0 \Rightarrow v = u$$

Which leads to $v \in U \Rightarrow (U^\perp)^\perp \subset U$.

$$\text{Cov. } U = V \Leftrightarrow U^\perp = \{0\}$$

Def. Orthogonal Projection 正交投影

Let U is a finite-dimensional subspace of V . The orthogonal projection of V onto U is the operator $P_U \in \mathcal{L}(V)$:

$$P_U v = u \quad \text{iff } v = u + w, \quad u \in U, \quad w \in U^\perp$$

P_U 把 v 拍到低维空间。

Prop. Suppose U is a finite-dimensional subspace of V .

a. $P_U \in \mathcal{L}(V)$

P_U 定义本身不显式说明，逐一验证。

b. $\forall u \in U, P_U(u) = u \quad \forall w \in U^\perp, P_U(w) = 0$

显然。

c. $\text{range } P_U = U \quad \text{null } P_U = U^\perp$

显然。

d. $v - P_U(v) \in U^\perp$

定义。

e. $P_U^2 = P_U$

If $v = u + w$, then $P_U v = u, P_U(P_U v) = P_U u = u$ from above.

f. $\forall v \in V, \|P_U v\| \leq \|v\|$

$$\|v\|^2 = \langle u + w, u + w \rangle = \langle u, u \rangle + \langle w, w \rangle + 2\langle u, w \rangle = \|P_U u\|^2 + \langle w, w \rangle$$

g. If $\{e_m\}$ is an orthogonal basis of U , then $\forall v \in V$,

$$P_U v = \sum \langle v, e_i \rangle e_i$$

前文已用过。

Thm. Riesz Representation Theorem

Suppose V is finite-dimensional. For each $v \in V$, define $\varphi_v \in V'$ by

$$\varphi_v(u) := \langle u, v \rangle \quad \forall u \in V$$

Then $v \mapsto \varphi_v$ is a one-to-one function from V onto V' . 双射

proof for surjective: a constructive proof.

If $\varphi = 0$, then $\varphi = \varphi_0$. Thus assume $\varphi \neq \varphi_0$, hence $\text{null } \varphi \neq V$, which implies $(\text{null } \varphi)^\perp \neq \{0\}$. Take $w \in (\text{null } \varphi)^\perp \wedge w \neq 0$. (From the conclusion, we know actually $w \parallel v$. Assume $w = kv$, then $\|w\|^2 = \langle kv, kv \rangle = k \langle v, kv \rangle = k \bar{k} \|v\|^2$, $\overline{\varphi(w)} = \overline{\langle kv, v \rangle} = \langle v, kv \rangle = \bar{k} \|v\|^2$) Let

$$v = \frac{\overline{\varphi(w)}}{\|w\|^2} w$$

Take the norm of both sides, $\|v\| = \frac{|\varphi(w)|}{\|w\|}$. Apply φ to both sides gives $\varphi(v) = \frac{\overline{\varphi(w)}\varphi(w)}{\|w\|^2} = \frac{|\varphi(w)|^2}{\|w\|^2} = \|v\|^2 = \langle v, v \rangle$. Now for each $u \in V$, rewrite it as

$$u = \left(u - \frac{\varphi(u)}{\varphi(v)} v \right) + \frac{\varphi(u)}{\varphi(v)} v$$

$$\varphi \left(u - \frac{\varphi(u)}{\varphi(v)} v \right) = \varphi(u) - \frac{\varphi(u)}{\varphi(v)} \varphi(v) = 0$$

$$\Rightarrow u - \frac{\varphi(u)}{\varphi(v)} v \in \text{null } \varphi$$

$$\stackrel{v \in \text{null } \varphi^\perp}{\Rightarrow} \langle u, v \rangle = \left\langle \frac{\varphi(u)}{\varphi(v)} v, v \right\rangle = \frac{\varphi(u)}{\varphi(v)} \|v\|^2 = \varphi(u)$$

2. Minimization Problems

Given a linear map T , we want to solve $Tx = y$. However, T may not be invertible, and y may not be in range T . With the help of orthogonal projection, we can still minimize $\|Tx - y\|$.

Prop. Minimizing distance to a subspace

Suppose U is a finite-dimensional subspace of V , $v \in V$, $u \in U$. Then

$$\|v - P_U v\| \leq \|v - u\|$$

The inequality is an equality iff $u = P_U v$.

proof. $v - P_U v \in U^\perp$, $P_U v - u \in U$. Therefore $\|v - P_U v\|^2 \leq \|v - P_U v\|^2 + \|P_U v - u\|^2 = \|v - P_U v + P_U v - u\|^2 = \|v - u\|^2$

Function Approximation

任取一组正交基用于拟合。必须指定拟合的区间，以确定内积定义。

$$\langle f, g \rangle := \int_l^r f(x)g(x) dx$$

$$\min \|f - g\| \Leftrightarrow \min \int_l^r (f - g)^2 dx$$

泰勒级数只保证邻域内精度，而此方法在完整定义域上最优。

- Power Series: need to do Gram-Schmidt procedure first
- Fourier Series

3. Pseudoinverse

最小化问题中把正交投影 $P_U y$ 当作 $Tx = y$ 的解，在此基础上继续削 T 而成为可逆映射，引出伪逆的概念。

Prop. Restriction of a linear map to obtain a one-to-one and onto map

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then $T|_{(\text{null } T)^\perp}$ is an injective map of $(\text{null } T)^\perp$ onto $\text{range } T$.

proof. injective: To prove $Tv = 0 \Leftrightarrow v = 0$. If $v \in (\text{null } T)^\perp \wedge Tv = 0$, then $v \in (\text{null } T) \cap (\text{null } T)^\perp \Rightarrow v = 0$

range is $\text{range } T$: Obviously $\text{range } T|_{(\text{null } T)^\perp} \subset \text{range } T$. To prove $\text{range } T \subset \text{range } T|_{(\text{null } T)^\perp}$, take $w \in \text{range } T$, then there exists $Tv = w$. Since $V = (\text{null } T) \oplus ((\text{null } T)^\perp)$, then we can rewrite $v = x + y$, where $x \in \text{null } T$, $y \in (\text{null } T)^\perp$.

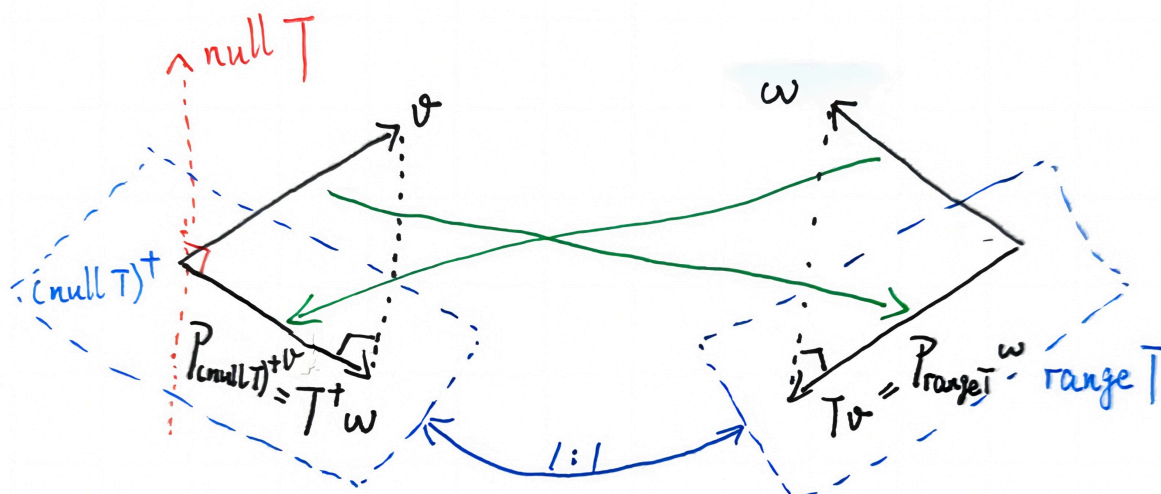
Hence, $T|_{(\text{null } T)^\perp} x = w$.

Therefore, $T|_{(\text{null } T)^\perp}$ is invertible.

Def. Pseudoinverse $T^\dagger$

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. The pseudoinverse $T^\dagger \in \mathcal{L}(W, V)$ of T is a linear map from W to V defined by

$$T^\dagger w := (T|_{(\text{null } T)^\perp})^{-1} \circ P_{\text{range } T} w$$



$\text{range } T \xleftrightarrow{1:1} V / \text{null } T$, 但 $w \in T$ 对应的是 $\text{null } T$ 的平移。伪逆将 w 对应到唯一的 $v \in V$, 并且（在内积的意义上） v 不含任何 $\text{null } T$ 的多余信息。

Prop. Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$

a. If T is invertible, then $T^\dagger = T^{-1}$

$\text{null } T = \{0\} \Rightarrow (\text{null } T)^\perp = V, \text{range } T = W$. Thus $T|_{(\text{null } T)^\perp} = T$ and $P_{\text{range } T}$ is the identity operator on W .

- b. $T \circ T^\dagger = P_{\text{range } T} : W \rightarrow W$, which is the orthogonal projection of W onto $\text{range } T$.
- c. $T^\dagger \circ T = P_{(\text{null } T)^\perp} : V \rightarrow V$, which is the orthogonal projection of V onto $(\text{null } T)^\perp$.

Prop. Pseudoinverse provides best approximate solution or best solution

Suppose V is finite-dimensional, $T \in \mathcal{L}(V, W)$, $w \in W$.

- a. If $v \in V$, then

$$\|T(T^\dagger w) - w\| \leq \|Tv - w\|$$

with equality iff $v \in T^\dagger w + \text{null } T$

- b. If $v \in T^\dagger w + \text{null } T$, then

$$\|T^\dagger w\| \leq \|v\|$$

with equality iff $v = T^\dagger w$